

Lab Experiment 5 : Linear Regression through Gradient Descent

Aim

To implement Linear Regression using the Gradient Descent optimization algorithm and evaluate its performance on a real world dataset.

Objectives

1. To understand the concept of Gradient Descent for optimizing a Linear Regression model.
2. To preprocess the dataset before model training.
3. To implement Linear Regression using Gradient Descent.
4. To analyze the convergence of the loss function during training.
5. To evaluate the model using standard regression metrics.

Dataset

Dataset Name: Student Performance Dataset

Source: UCI Machine Learning Repository

Dataset Link:

<https://archive.ics.uci.edu/dataset/320/student+performance>

Tasks to Perform

1. Download and load the Student Performance dataset.
2. Perform data preprocessing, including handling missing values (if any), encoding categorical variables, and feature scaling.
3. Select appropriate input features and the target variable.
4. Split the dataset into training and testing sets.
5. Implement Linear Regression using the Gradient Descent algorithm.
6. Experiment with different learning rates and observe their effect on model convergence.
7. Plot the loss (cost) versus the number of iterations.
8. Evaluate the trained model using:
 - Mean Absolute Error (MAE)
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score

9. Interpret the results and comment on the convergence behavior and prediction performance of the model.

Expected Learning Outcomes

Upon successful completion of this experiment, students will be able to:

1. Explain the working of the Gradient Descent optimization algorithm.
2. Implement Linear Regression using Gradient Descent.
3. Analyze the effect of learning rate on model convergence.
4. Evaluate regression models using standard performance metrics.
5. Interpret regression results and assess model performance.